

Tool	-@	Records	Mapq 60	Index (s)	Map (s)	Mem (GB)	Agree
shmap-rs	1	149 194	139 309	6.52	36.27	2.66	—
shmap-rs	4	149 194	139 309	3.06	9.19	2.03	—
shmap-rs	8	149 194	139 309	1.67	4.54	2.27	—
shmap-rs	16	149 194	139 309	1.39	2.30	2.14	—
shmap-rs	32	149 194	139 309	1.46	1.38	2.43	—
midrule cpp-shmap	1	149 194	139 309	25.42	77.67	18.85	—
midrule winnowmap2	32	189 022	—	—	1493.40	9.83	0.9633

Input: 149438 reads. Records are PAF lines, not reads: a tool emitting secondary alignments writes more than one per read.

Table 1: shmap-rs beside other long-read mappers on the headline real-HiFi dataset, same host, same reads. The thread column is part of the measurement: the peers run at the count their own documentation assumes, shmap-rs at one. *Agreement* is the share of that tool’s mappings shmap-rs places compatibly — concordance, not accuracy: where they differ, nothing here says which is right.

Generated benchmark artifacts

Typeset from `paper/generated/aarch64/` by `benchmarks/scripts/build_pdf.py`. Every number resolves to a row in the result set named below; see `PROVENANCE.md` for the inputs, transformation and caveats of each artifact.

```

result set: current
commit: cc90fa548767
host: galaxy
measured: 2026-08-10
inputs: sha256:a15b1477d0efdc6b

```

Dataset	Tool	Metric	-@	Mapped	Mapq 60	Missed (%)	Wrong Q60	Index (s)	Map (s)	Mem (GB)
B01	shmap-rs	Containment	1	149 194	139 309	6.78	—	6.52	36.27	2.66
B01	shmap-rs	Jaccard	1	146 120	138 421	7.37	—	6.50	50.77	2.65
B01	shmap-rs	bucket_SH	1	149 236	138 250	7.49	—	6.53	27.77	2.63
B01	cpp-shmap	Containment	1	149 194	139 309	6.78	—	25.42	77.67	18.85
B01	cpp-shmap	Jaccard	1	146 120	138 421	7.37	—	25.34	105.03	18.85
B01	cpp-shmap	bucket_SH	1	149 236	138 250	7.49	—	25.64	61.44	18.85
B01	winnowmap2	—	32	189 022	—	—	—	—	1493.40	9.83
B02	shmap-rs	Containment	1	125 000	117 532	5.97	0	6.54	27.82	2.66
B02	shmap-rs	Jaccard	1	125 000	119 065	4.75	6	6.50	43.53	2.65
B02	shmap-rs	bucket_SH	1	125 000	116 800	6.56	1	6.54	23.97	2.66
B02	cpp-shmap	Containment	1	125 000	117 532	5.97	—	25.60	60.76	18.85
B02	cpp-shmap	Jaccard	1	125 000	119 065	4.75	—	25.08	90.63	18.85
B02	cpp-shmap	bucket_SH	1	125 000	116 802	6.56	—	25.12	54.63	18.85
B02	winnowmap2	—	32	129 488	—	—	—	—	1123.80	9.47
B03	shmap-rs	Containment	1	241 991	227 521	6.19	—	6.49	45.23	2.65
B03	shmap-rs	Jaccard	1	241 265	228 887	5.63	—	6.53	61.01	2.65
B03	shmap-rs	bucket_SH	1	242 050	226 931	6.43	—	6.51	36.23	2.65
B03	cpp-shmap	Containment	1	241 991	227 521	6.19	—	25.57	86.53	18.85
B03	cpp-shmap	Jaccard	1	241 265	228 886	5.63	—	25.68	117.10	18.85
B03	cpp-shmap	bucket_SH	1	242 050	226 934	6.43	—	26.06	69.73	18.85
B03	winnowmap2	—	32	280 093	—	—	—	—	1113.10	8.22
B04	shmap-rs	Containment	1	2 419 796	2 273 122	6.28	—	6.54	455.45	2.66
B04	shmap-rs	Jaccard	1	2 411 461	2 286 085	5.74	—	6.52	622.92	2.65
B04	shmap-rs	bucket_SH	1	2 420 331	2 267 229	6.52	—	6.51	360.70	2.65
B04	cpp-shmap	Containment	1	2 419 796	2 273 121	6.28	—	25.04	877.05	18.85
B04	cpp-shmap	Jaccard	1	2 411 461	2 286 083	5.74	—	25.12	1195.37	18.85
B04	cpp-shmap	bucket_SH	1	2 420 331	2 267 238	6.52	—	25.12	703.13	18.85
B04	winnowmap2	—	32	2 808 979	—	—	—	—	10434.70	8.47
B05	shmap-rs	Containment	1	39 608	37 839	58.97	—	6.51	13.68	2.65
B05	shmap-rs	Jaccard	1	5 956	5 409	94.13	—	6.54	16.80	2.66
B05	shmap-rs	bucket_SH	1	41 115	39 226	57.46	—	6.49	12.57	2.65
B05	cpp-shmap	Containment	1	39 608	37 839	58.97	—	25.18	26.58	18.85
B05	cpp-shmap	Jaccard	1	5 956	5 409	94.13	—	25.15	31.18	18.85
B05	cpp-shmap	bucket_SH	1	41 115	39 225	57.47	—	25.21	25.02	18.84
B05	winnowmap2	—	32	99 107	—	—	—	—	327.10	6.11

Table 2: Long-read mapping tools on the benchmark suite. Mapq 60 is the count of confident mappings; Missed is the share of input reads either unmapped or below mapq 60; Wrong Q60 counts confident mappings that are wrong against ground truth.

Dataset	Metric	Matches per read			Potential		Buckets per read	
		possible	examined	in mapping	realized	unrealized	seeded	final

Table 3: Efficiency of the seed heuristic. Realized potential is possible matches divided by matches examined (work avoided); unrealized potential is matches examined divided by matches inside the reported mapping (work still wasted).

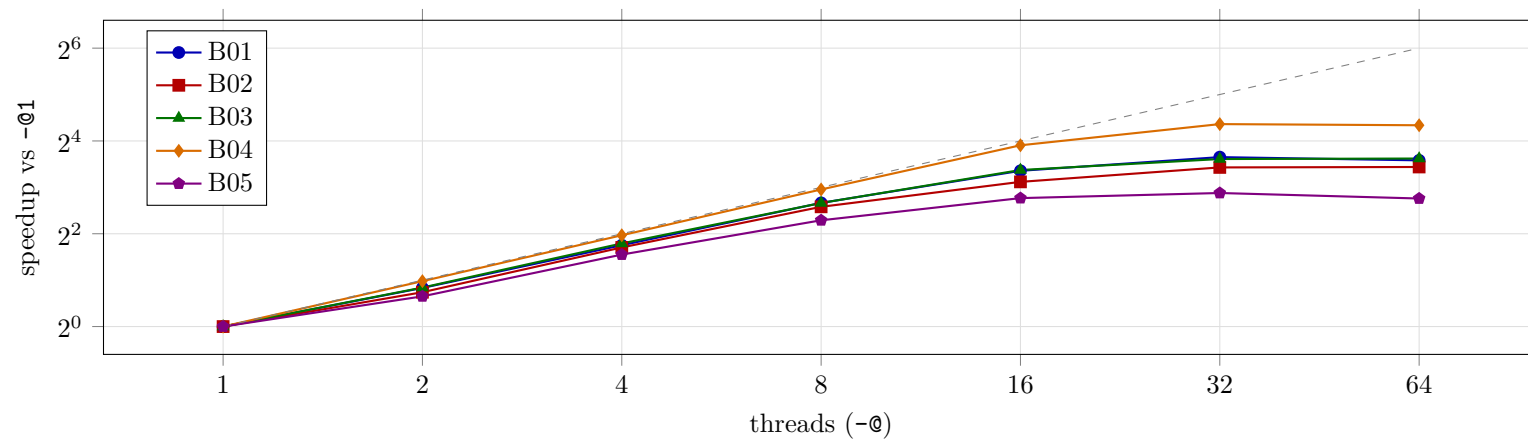


Figure 1: Whole-run speedup against thread count, Containment. The dashed line is linear speedup.

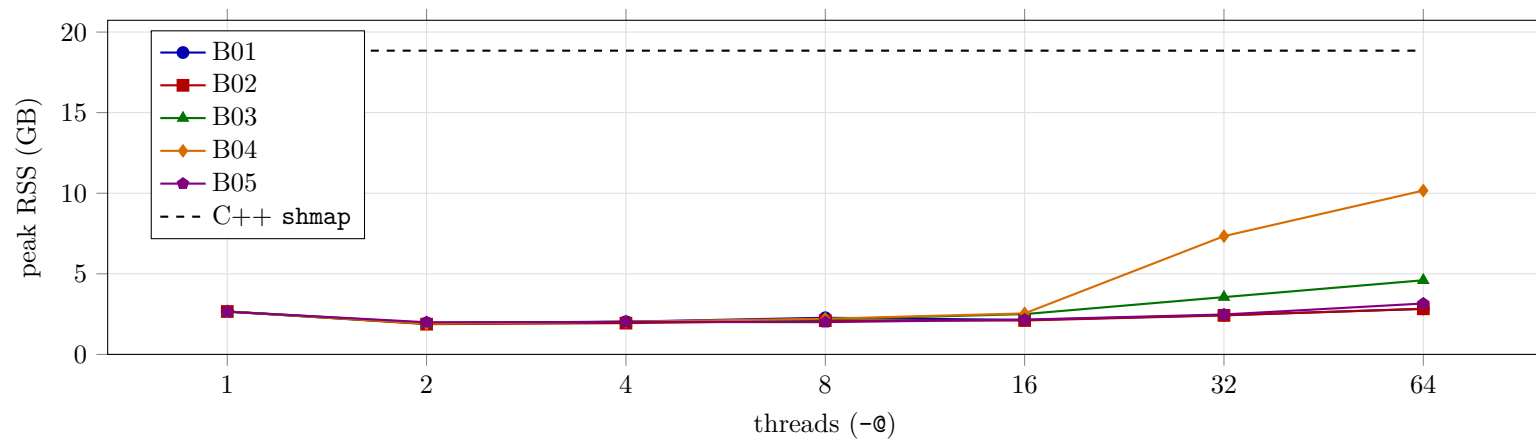


Figure 2: Peak resident memory against thread count, Containment, with the C++ reference as a horizontal line.

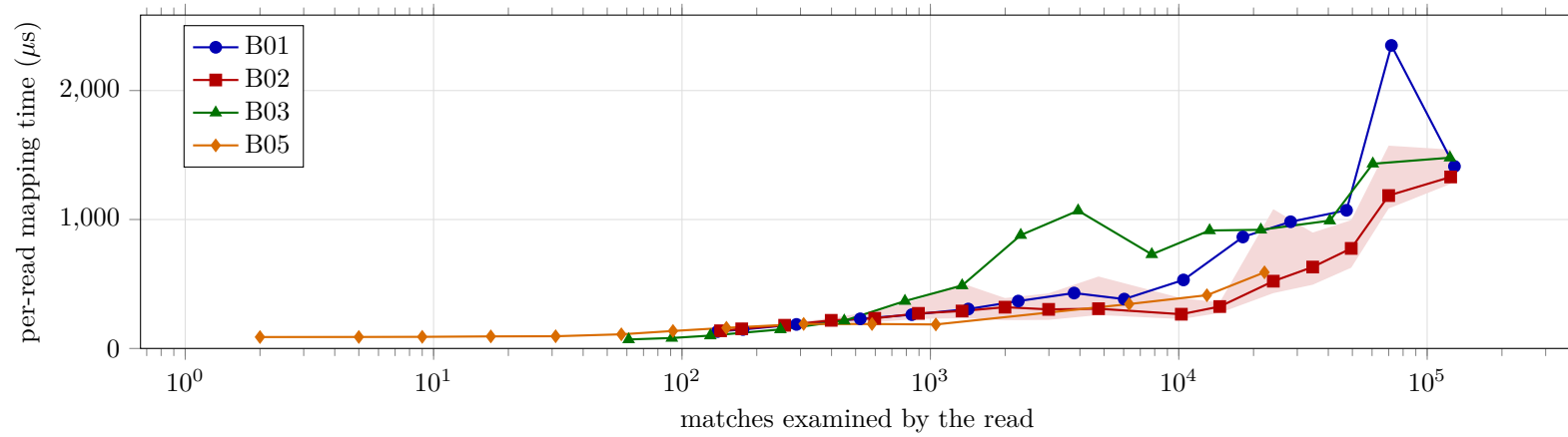


Figure 3: Per-read mapping time against the number of matches the read examined, Containment. Points are per-bin medians over reads grouped into 18 logarithmic bins; the shaded band is the interquartile range for the simulated set.

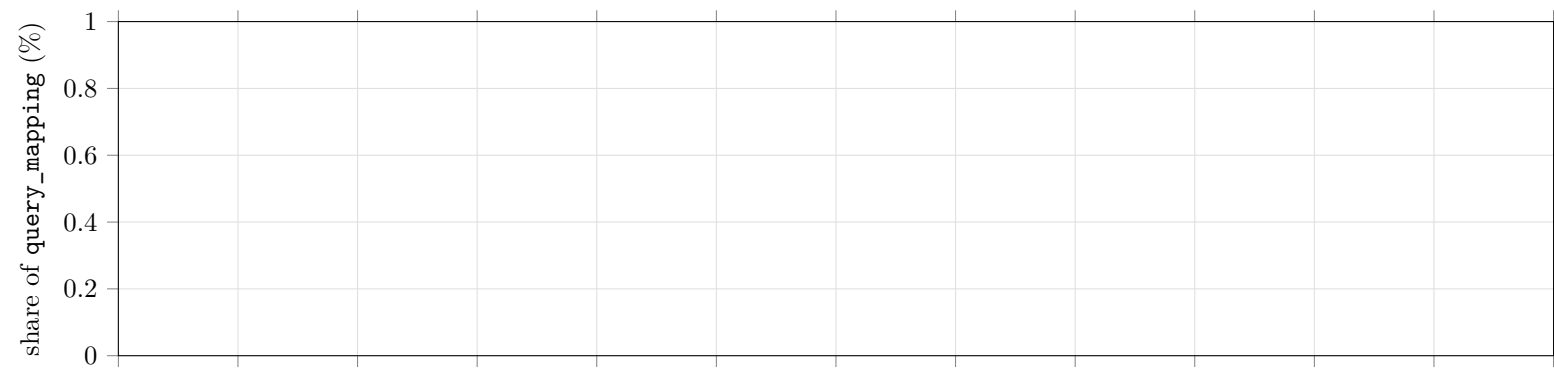


Figure 4: Where mapping time goes: stage shares of `query_mapping`, single-threaded.